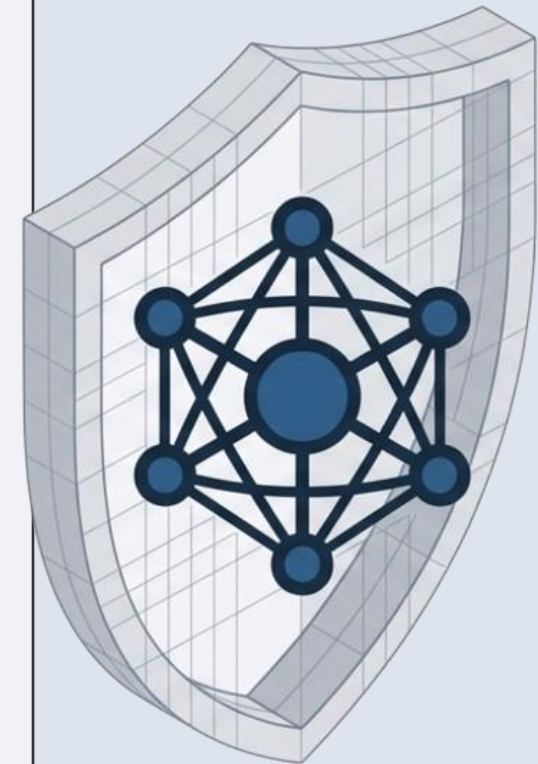


Cyber Threat Intelligence (CTI) Platform for Open-RAN

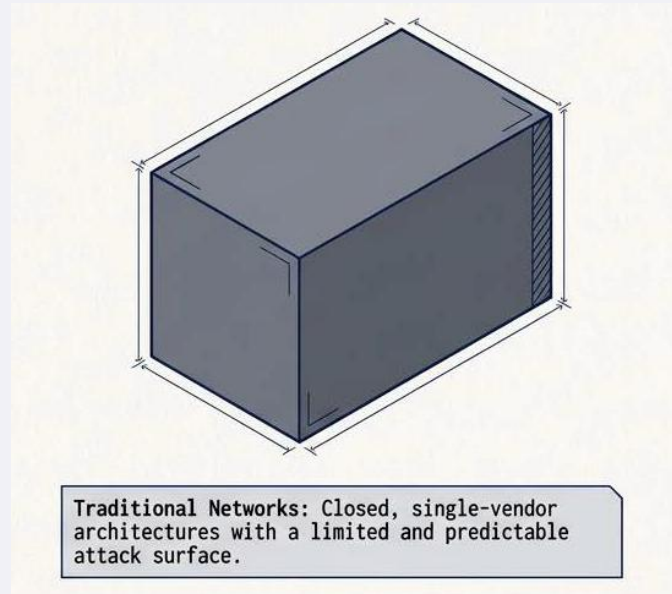
H.G.R. Achintha	EG/2021/4384
I.U. Madakaladeniya	EG/2021/4651
P.A.G.N. Rathnaweera	EG/2021/4751
S.A. Wijesinghe	EG/2021/4877

Supervisor : Dr. Geeth Priyankara

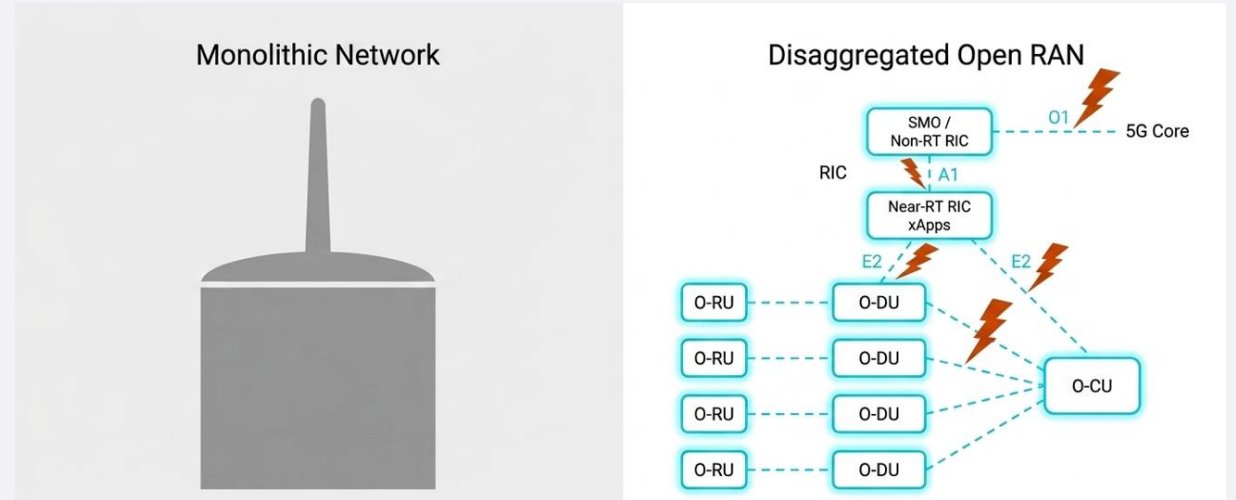
Co-Supervisor : Assistant Prof. Pasika Ranaweera



The O-RAN Paradigm - Flexibility vs. Vulnerability



Traditional networks: closed, single-vendor architectures with a limited and predictable attack surface.



O-RAN: unprecedented flexibility through interoperable components, completely redefining the security perimeter.

O-RAN Specific CTI Absence

Existing enterprise CTI tools are blind to specific O-RAN components (O-RU, O-DU, O-CU).

Lack of RIC Integration

Security fails to natively use the RAN Intelligent Controller for programmable defence.

Insufficient Threat Sharing

No standardised, privacy-preserving protocol, creating reactive silos not collaborative defence.

E2E Detection & Mitigation

No end-to-end threat detection and mitigation with CTI in the context of O-RAN.

Solution

A disaggregated CTI platform on a native O-RAN with autonomous threat dissemination

O-RAN Domain-Native CTI & STIX 2.1 Graph

Synthesises telecom-specific threat intelligence across all 10 O-RAN components, open interfaces (E2, A1, O1) and multi-vendor radio attack paths.

Max-Sim semantic tensor pruning of MITRE ATT&CK and FIGHT, ArangoDB STIX 2.1 graph store (:8529), OASIS TAXII 2.1 REST server (:5001)

Dual-Tier RIC Integration (xApps + rApps)

Natively integrates detection, vector retrieval and tactical defence directly into Near-RT and Non-RT RIC control loops with locality-first sub-second edge resolution.

xApp2 telemetry ingestion, all-MiniLM-L6-v2 ChromaDB vector store with strict threshold gate ($t_L = 0.69$) for <50 ms local response, Gemini 1.5 Pro with OpenRouter fallback.

Inter-Platform Agent & FastMCP Exchange

Enables collaborative cross-region threat defence by securely synchronising intelligence and dynamic edge learning without exposing operational privacy or creating broadcast loops.

FastMCP tool escalation (:8000/mcp), automated PII and IP scrubbing, source-exclusion loop prevention, local ChromaDB PVC upsert.

Deterministic Gateway & Guarded Mitigation

Provides end-to-end protection by grounding multi-modal anomalies in verifiable mathematical metrics and executing safety-gated, reversible network responses.

Multi-modal E2SM-KPM, Falco and Loki V% scoring, xApp1 E2SM-RC throttle and CNI pod quarantine, PromQL closed-loop effect verification with automated safe rollback.

Initial Project Objectives

01

O1: Design and Implement a Disaggregated, O-RAN-specific CTI Platform

- Build multi-region testbed with Near-RT RIC, srsRAN/Open5GS, xApps and regional intelligence.
- Collect security evidence from O-RAN and cloud sources (E2SM-KPM, Prometheus, Loki, Falco).
- Ingest ATT&CK, FiGHT and vulnerabilities into O-RAN-aware CTI registries.
- Enable prioritised, privacy-aware dissemination with provenance and dashboard.

02

O2: Integrate Agentic AI for O-RAN Threat Contextualisation & Response

- Develop regional RAG agents for CTI retrieval and severity classification.
- Coordinate escalation to Global-RAG via MCP when regional evidence is insufficient.
- Return global knowledge to the region, control growth via canonicalisation and de-duplication.
- Generate contextual recommendations with policy gates, approval, verification and rollback.

03

O3: Validate and Evaluate the O-RAN CTI Platform & Agentic AI Workflow

- Measure retrieval, contextualisation and knowledge growth against an OpenCTI baseline.
- Validate detector capture and evidence transparency with controlled scenarios.
- Evaluate bounded mitigation: execution, verification, timing and rollback.
- Observe operational health via the security score and component metrics.

Extended Project Objectives

Two major capabilities were added as the project evolved from the progress-report proposal into the final multi-region platform.

Implement the Anomaly Detection Gateway for O-RAN Telemetry

- Ingest and correlate multi-modal telemetry across O-RAN radio, container runtime syscalls and control-plane logs.
- Execute deterministic anomaly scoring and baseline evaluation using rolling 3-sigma statistical thresholds.
- Record immutable, normalised ThreatEvent ledgers in rApp2, preserving affected component, interface and structured rule_evaluations.
- Validate detector transparency and evidence contracts across controlled synthetic scenarios.

Set Up Automated Mitigation Pipeline for Closed-Loop O-RAN Defence

- Enforce safety policy gates and allow-listed actuators (xApp1 E2SM-RC throttle, A1 policy, pod quarantine) with explicit critical-infrastructure protection.
- Implement closed-loop PromQL telemetry verification across a 5-stage state machine to confirm security outcomes.
- Execute automated safe rollback upon verification failure or timeout, restoring baseline radio policies and network connectivity.

System Architecture

1 Threat intelligence sources

MITRE ATT&CK, MITRE FIGHT and vulnerability feeds enter through the Threat Data Ingestion rApp.

2 Non-RT RIC / O-Cloud / SMO

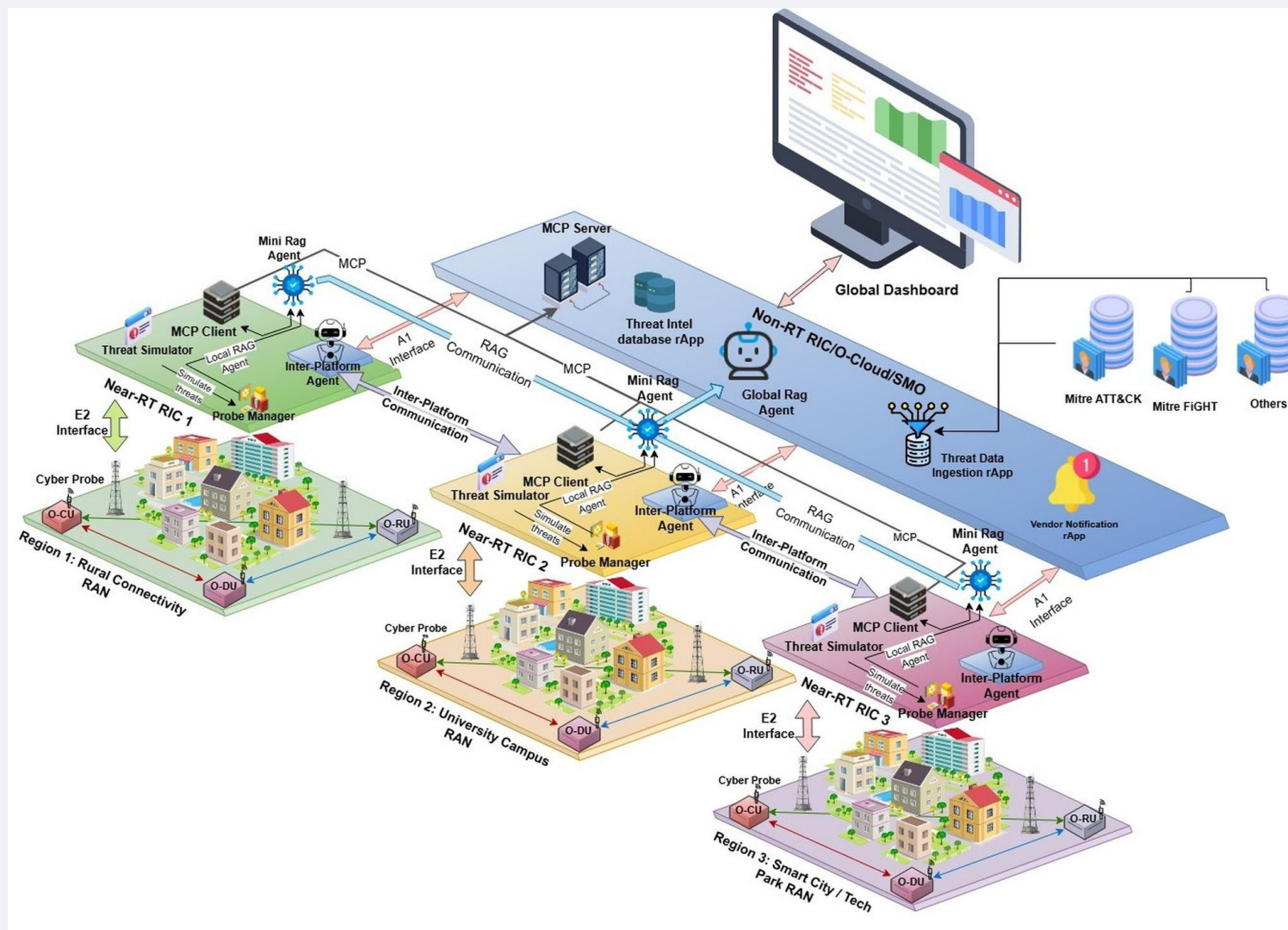
Global RAG agent, MCP server, threat-intel database rApp, vendor notification rApp and the global dashboard.

3 Near-RT RIC regions

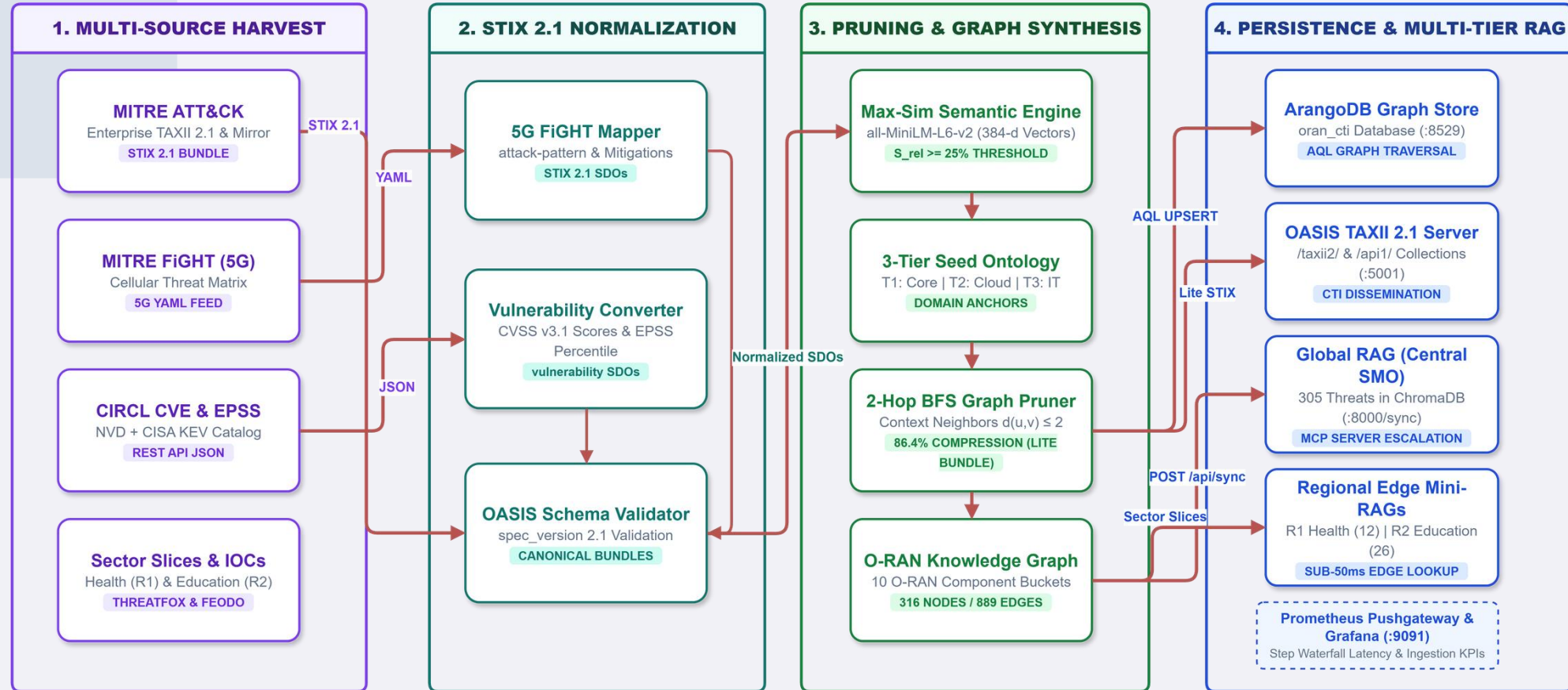
Each region runs a Threat Simulator, Probe Manager, MCP client, Mini RAG agent and Inter-Platform Agent.

4 RAN and radio layer

srsRAN O-CU, O-DU and O-RU with cyber probes, observed by the region over the E2 interface.



Threat Ingestion & Normalization Pipeline



- Semantic Tensor Pruning - dual-stage Max-Sim filtering prunes 85 to 90% of irrelevant enterprise IT noise from MITRE ATT&CK and FiGHT, retaining only O-RAN-specific threat vectors.
- Domain-Native TAXII server & STIX 2.1 Graph - combined threats across all 10 O-RAN architectural components into an ArangoDB graph database (:8529), exposed via an OASIS TAXII 2.1 server (:5001).

Regional Mini-RAG and Global-RAG Workflow

1. Locality-First Decision Gate

Local Answer ONLY when Thresholds Pass

- **Strict Gate Rule:**
 $s_1 \geq \tau_L (0.69) \ \& \ E_L = 1$
- **Edge Vector Retrieval:**
Queries regional ChromaDB store K_r (1,679+ threat records) using *all-MiniLM-L6-v2* (384-d).
- **Raw Cosine Similarity:**
Calculates exact uninflated score $s_1 = \max s(q, d)$ with component/interface re-ranking.
- **Operational Latency:**
Provides local response in **< 50 ms** at zero external cloud LLM API cost.

2. FastMCP Global Escalation

Escalation via Standard FastMCP Protocol

- **Escalation Trigger:**
Dispatches when $s_1 < 0.69$ or $E_L = 0$ (insufficient local evidence).
- **FastMCP Tool Protocol:**
Calls tool `mini_rag.analyze_threat` on Central SMO (:8000/mcp).
- **Standardised Payload Carries:**
 1. Normalised *ThreatEvent* & deterministic rule facts
 2. Regional candidate matches & peak score s_1
 3. Explicit escalation reason.
- **Central STIX Knowledge:**
Queries **305 consolidated threats** across 10 O-RAN components + **ArangoDB STIX 2.1 Graph** (:8529).

3. Deterministic LLM Fallback

Strictly Ordered LLM Fallback Chain

- **Tier 1 (Primary Provider):**
Google Gemini 1.5 Pro executes structured analytical reasoning.
- **Tier 2 (Fallback Provider):**
OpenRouter API engages upon HTTP 429 quota or 503 outage.
- **Tier 3 (Explicit Error State):**
Generation Error State emitted if both fail.
- **Zero FalseText Guarantee:**
Never outputs silent handcrafted text. Detector telemetry facts (**V%**) are 100% preserved without modification.

4. Standardised 7-Section Report

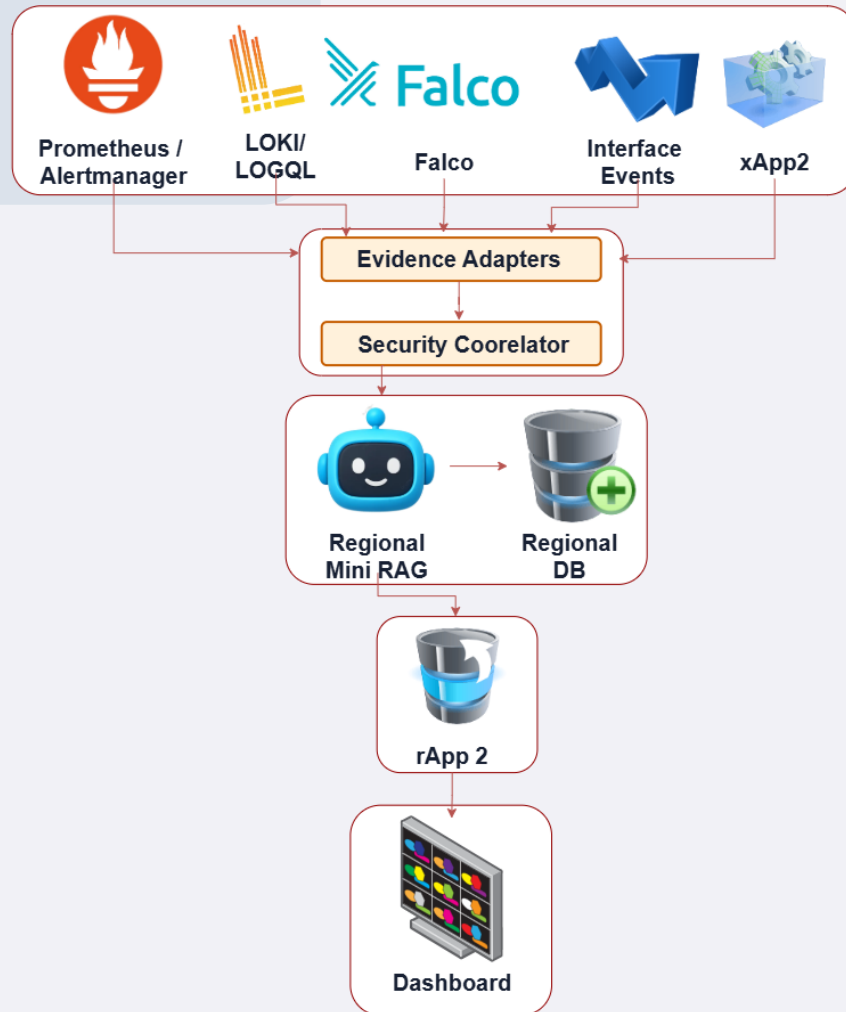
Returned for Every Accepted Analysis

1. **Threat Identification:**
Incident class, affected entity, pod/UE, interface.
2. **Attack Pattern & Kill Chain:**
Attacker behaviour, MITRE ATT&CK / FiGHT stage.
3. **O-RAN Impact Assessment:**
RAN, Near-RT RIC, xApps & O-Cloud impact.
4. **Evidence Correlation:**
Direct link to fired detector rules & telemetry V%.
5. **Immediate Actions:**
Bounded containment & investigation steps.
6. **Detailed Mitigations:**
Long-term remediation & architectural hardening.
7. **Severity Assessment:**
LOW / MEDIUM / HIGH / CRITICAL with rationale.



- Escalation trigger - Global-RAG is engaged only when local similarity drops below the threshold.
- Learning criteria - integration is limited to valid, usable and novel records; existing families are ignored.
- The RL metric - growth efficiency is learned records over eligible escalations, not raw threat volume.
- Decision separation - escalation and learning are deliberately decoupled.

Anomaly Detection Gateway



Multi-Modal Telemetry Streams

- Radio metrics (E2SM-KPM / xApp2)
- Container runtime (Falco)
- Control-plane logs (Loki)
- Interface events and Prometheus alerts

Deterministic Scoring Engine

- Direct threshold and 3-sigma baseline evaluation
- Correlation over a 300 s window
- Zero AI involvement in the detection decision

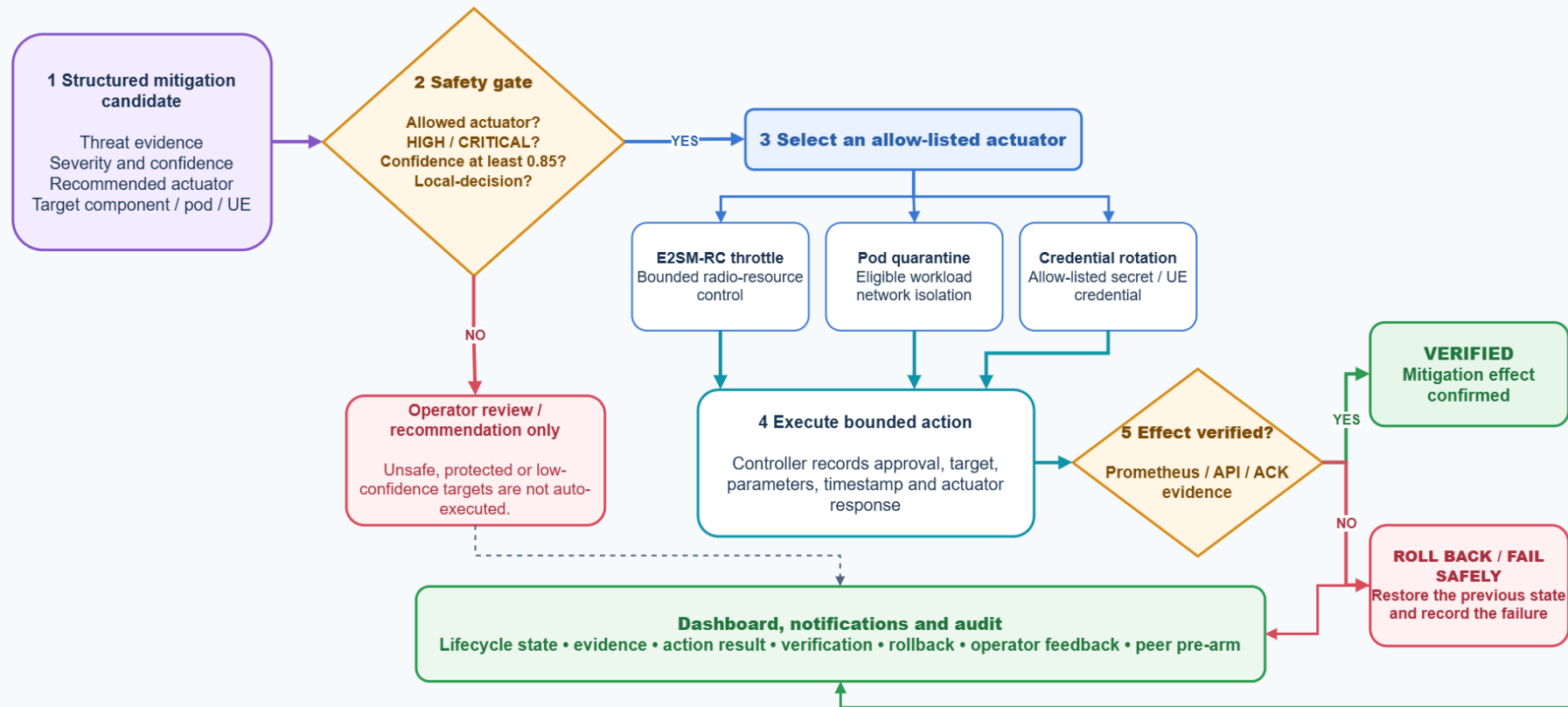
Normalized ThreatEvent Recording

- Standardised JSON contract
- Rule, criterion, measured value and threshold retained
- Immutable persistence in rApp2

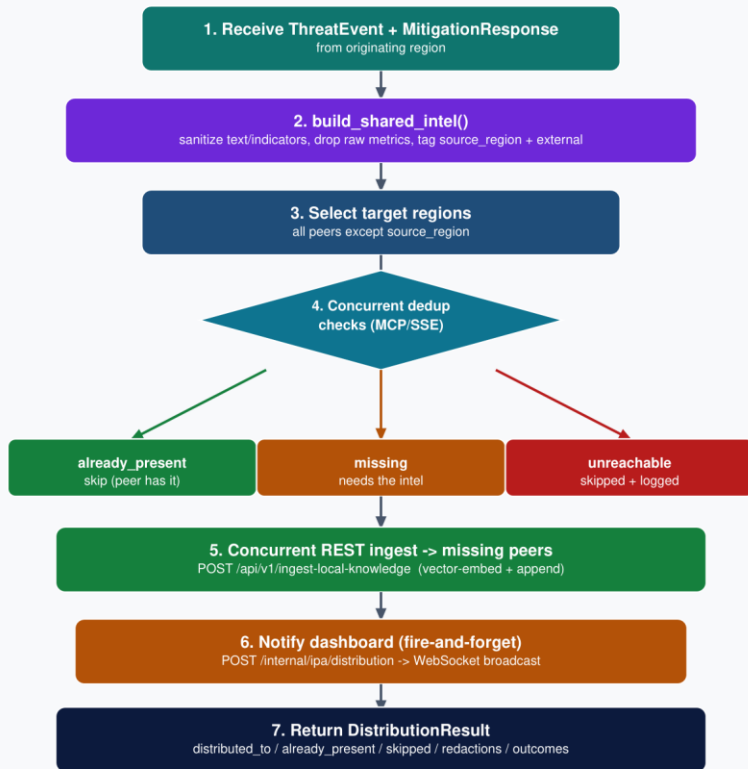
Anomaly Families - Emulation and Detection

Family	Mechanism (bounded / labelled simulation)	Expected detector family
UE DoS / KPM	Bounded iperf load + KPM throughput evidence	KPM threshold / baseline
UE runtime shell	Bounded shell execution with experiment marker	Falco container shell rule
A1 anomaly (simulated)	Structured invalid / unauthorised policy burst	A1 policy validation failure
F1 anomaly (simulated)	Structured F1 interface anomaly contract	F1 interface anomaly family
E2 signalling (simulated)	Structured E2 signalling anomaly contract	E2 signalling / subscription family
xApp invalid API	Bounded invalid threat-event API requests	OranXappInvalidApiBurst
Pod runtime shell	Short-lived Alpine pod shell marker	Falco shell-spawn family
Memory pressure	Resource-limited workload, sustained pressure	Memory limit pressure family
Auth log burst	Controlled authentication failure log messages	Region auth failure burst

Guarded Closed-Loop Mitigation & Actuation



Inter-Platform Agent - Privacy-Preserving Sharing



Sanitise before it leaves

`build_shared_intel()` scrubs indicators and raw metrics, drops operational identifiers and tags the source region as external intel.

Source exclusion prevents loops

The originating region is removed from the target set, so shared intelligence can never be reflected back to its author.

Concurrent dedup, not broadcast

Every remaining peer is asked over MCP/SSE whether it already holds the intel. Peers answer `already_present`, `missing` or `unreachable`.

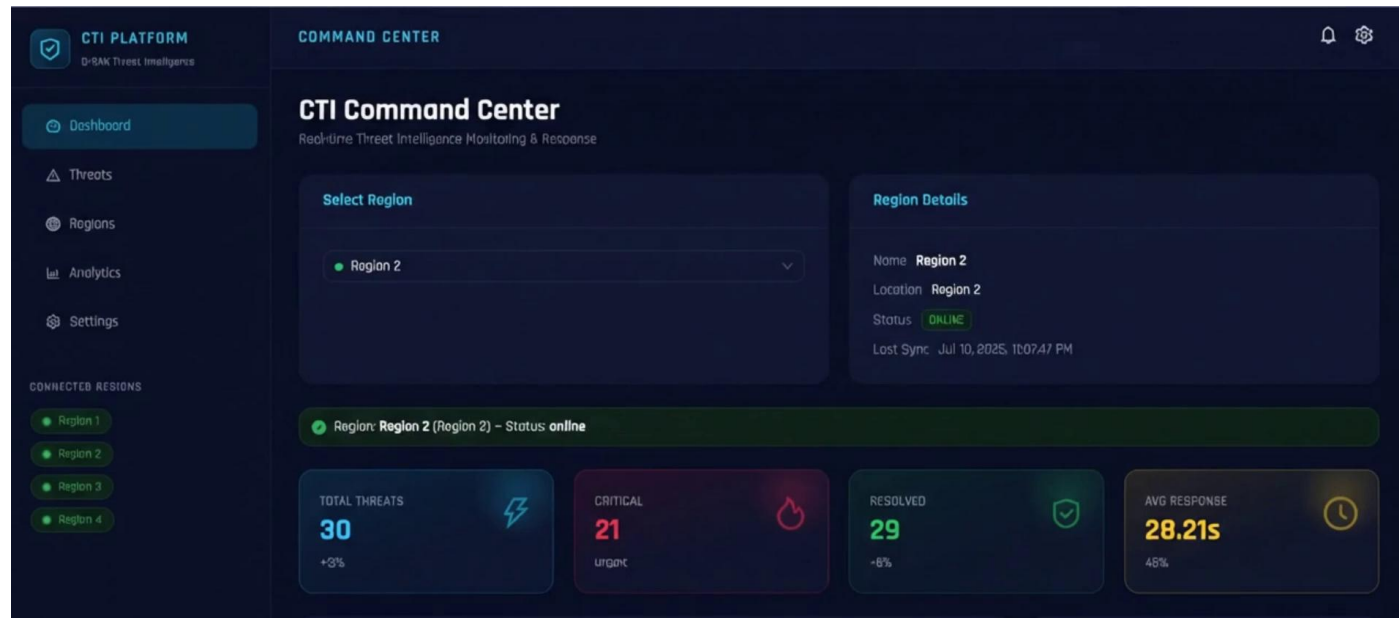
Selective REST ingest

Only peers that lack it receive a POST to `/api/v1/ingest-local-knowledge`, which vector-embeds and appends into that region alone.

Auditable outcome

`DistributionResult` records `distributed_to`, `already_present`, `skipped` and the redaction counts for every attempt.

Operator Command Dashboard



Command centre: region selector, live status and per-region threat counters

Live deployment: <http://193.1.132.238/>

Region-scoped single pane

One view over all four regions with a selector; each region streams its own live updates over WebSocket.

Evidence transparency

Every threat card exposes the detector rule, measured value and threshold, plus the matched knowledge-base records and their similarity.

Mitigation history

Full candidate lifecycle with verification and rollback state, persisted rather than live-only.

Intelligence growth tab

Session scores, metric composition and weight-sensitivity, served from /cti/intelligence/growth.

Subscriber Notifications & Operator Feedback



Ishini Madakaladeniya <uppalee@gmail.com>

O-RAN CTI alert for region_2 (LOW)

2 messages

projectdeploying@gmail.com <projectdeploying@gmail.com>
Reply-To: projectdeploying@gmail.com
To: uppalee@gmail.com

Sun, Jul 26, 2026 at 5:50 PM

O-RAN CTI ALERT LOW Threat — region_2

Detected Threat Summary

O-RAN workload memory is outside its rolling baseline. Rule 'Rolling memory-baseline anomaly' measured 1.002 baseline ratio; the trigger condition is > 1 baseline ratio, exceeding the threshold by 0.20%.

Affected O-RAN Component: mini-rag-agent
Affected Interface: A1
Severity Level: LOW
Confidence Score: 58%
Source Region & Sharing: Locally detected (not shared)

Take Action Now (Immediate)

- Increase monitoring for the affected O-RAN interface, preserve the telemetry window, and wait for corroborating indicators before containment.
- Isolate the affected 'mini-rag-agent' component.
- Increase monitoring and logging for the 'mini-rag-agent' and its associated network traffic.
- Temporarily disable any recently deployed or modified policies related to the 'mini-rag-agent'.

Detailed Mitigations

- Investigate telemetry, logs, and configuration changes in O-Cloud and RAN-General for indicators related to the 2022 Ukraine Electric Power Attack (C0034), Andarief (G0138), Exclusive Control (T1668), ToddyCat (G1022), Masquerading (FGT1036), Kitten (S0278), and Sys10 (S0060).
- Apply least-privilege access, service hardening, patching, and configuration validation for the affected O-RAN component ('mini-rag-agent').
- Correlate the detection with recent alarms, interface traffic, authentication events, and component health metrics before triggering automated response.
- Ensure robust resource management and monitoring for O-RAN pods to detect and alert on memory baseline anomalies.
- Implement regular security audits and vulnerability assessments for O-Cloud and RAN-General components.

RAG-Generated Mitigation Report

<https://mail.google.com/mail/u/0/?ik=54c9523e3&view=pt&search=all&permthid=bread-f1871779779207100368&siml=msg-f1871779779207100368> 1/8

Delivered Region 2 alert

Operator Mitigation Record

Authentication failure burst

Region-1 authentication failure burst. Rule 'Authentication failure burst' measured 302 log matches; the trigger condition is >= 6 log matches, exceeding the threshold by 4933.33%.

Automated mitigation executed unverified

Credential rotation

on secret demo-managed-credential - key token mit-24f93e3f523141c8

Pending

Approved

Executing

Executed

Verifying

Verified

Executed Unverified

Latest usefulness analysis

0 of 23 generated actions were used (0.0% coverage); 1 extra operator actions recorded.

Operator Email

uppalee@gmail.com

Mitigation Status

Pending

Immediate Actions

Increase monitoring for the affected O-RAN interface, preserve the telemetry window, and wait for corroborating indicators before containment.

Investigate telemetry, logs, and configuration changes in SMD, RAN-General, O-Cloud for indicators related to Indicator Blocking.

Investigate telemetry, logs, and configuration changes in O-Cloud for indicators related to Container Administration Command.

Detailed / Follow-up Actions

Apply Multi-factor Authentication (M1032) controls relevant to Cloud Services.

Apply Privileged Account Management (M1026) controls relevant to Cloud Services.

Prioritize evidence terms used in the mapping: authorization, OAuth, private cloud.

Investigate telemetry, logs, and configuration changes in IAM for indicators related to Local Accounts.

Apply least-privilege access, service hardening, patching, and configuration validation for the affected O-RAN

Operator action and knowledge-feedback capture

Subscription model

Vendors, operators and security teams subscribe per region; an alert is sent when a threat is detected, analysed, mitigated or shared.

Emailed once, never twice

A notification log keyed per threat guards idempotency, so a re-broadcast or a restart cannot spam a subscriber.

Immediate actions first

The message leads with the take-action-now tier, then detailed mitigations, evidence correlation and severity rationale.

Feedback closes the loop

Applied actions, extra actions and a usefulness rating are persisted with the threat and can become reviewed regional knowledge.

14

Testbed Deployment & Experimental Evaluation

64 Physical Cores
(128 Logical)

250 GB Host
RAM

K3s v1.36.2
Single Node Control Plane

4 Regions
One Node, 4 Namespaces

AMD Ryzen Threadripper PRO 7985WX host with Ubuntu 24.04.4 LTS, containerd 2.3.2-k3s2

Region
1

- 2 gNB groups
- 6 UEs
- O-RAN SC RIC
- xApps
- Mini RAG agent

Region
2

Region
3

- Sharing peers
- Receive distributed intel
- Pre-armed for cross-region response

Region
4

8-point readiness checklist run before every campaign

namespace health, correlator to Mini-RAG reachability, rApp2 routes, Gemini key fingerprinting, pre-campaign knowledge snapshot.

Evaluation Design

The 750-event campaign

A deterministic 150-slot schedule run over five days across two evaluated regions, spanning eight threat families: UE DoS and KPM anomalies, runtime shell, A1, F1 and E2 interface anomalies, xApp API abuse, pod shell and memory pressure.

Frozen intelligence benchmark

A fixed, checksummed ground-truth case set evaluated read-only, so scores are comparable across sessions. It measures grounded retrieval recall, ranking, local-versus-global routing, severity accuracy and generation reliability.

The declared composite score

54%

RAG intelligence

retrieval, ranking, routing, severity, generation

36%

End-to-end detection

persisted event capture and expected-rule accuracy

10%

Mitigation response

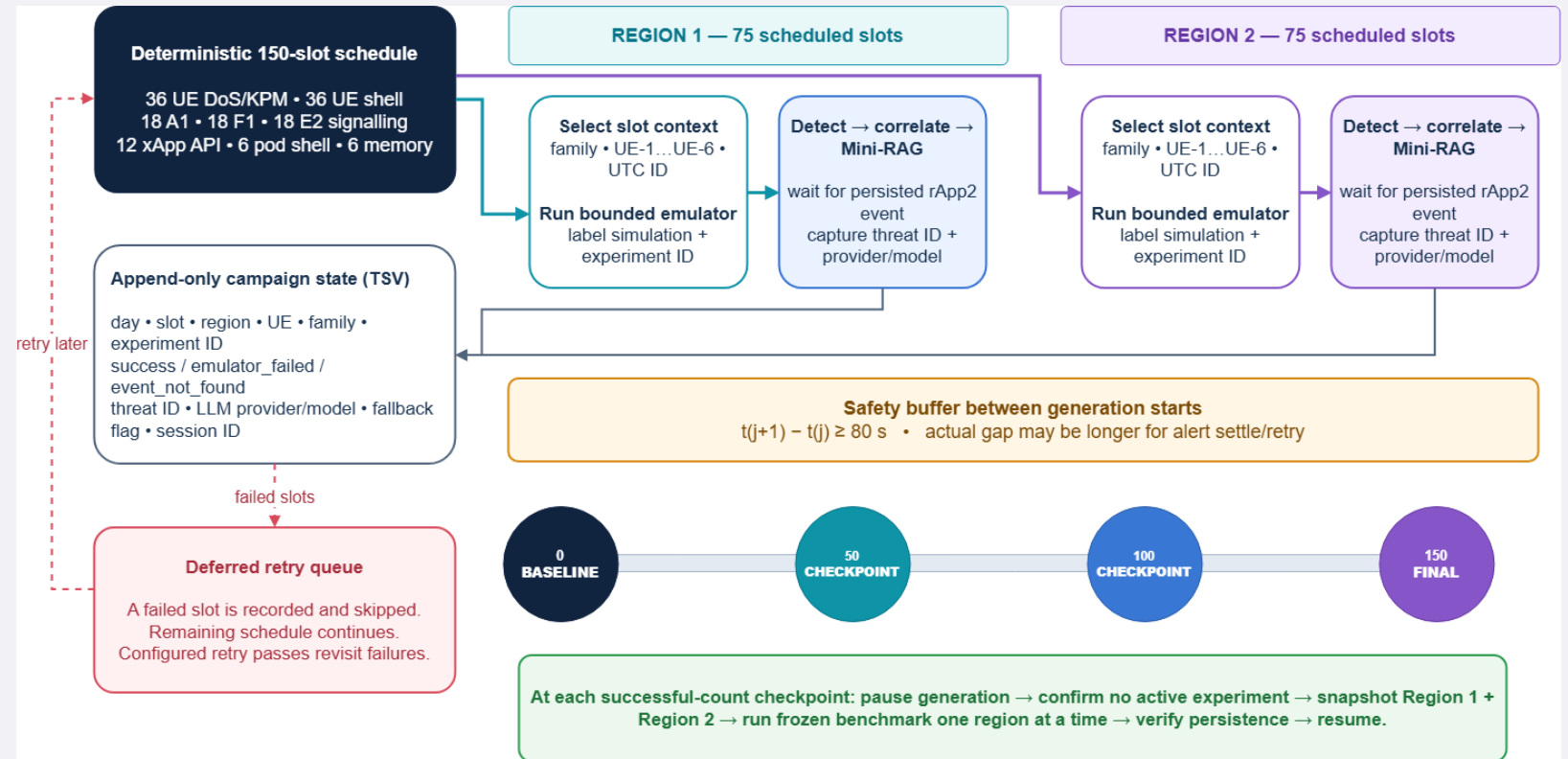
execution, verification, timeliness, cross-region gain

Weights are pre-declared, never learned. A component that was not measured is omitted and the remainder renormalised, never scored as zero and never assumed successful.

Evaluation - 150-Event Anomaly Campaign

Declared 150-slot composition
(75 slots per region)

UE DoS / KPM	36	24%
UE runtime shell	36	24%
A1 / F1 / E2 anomaly	54	36%
xApp invalid API	12	8%
Pod shell / memory	12	8%



- Run for 5 days -> 750 events; a new campaign ID and knowledge snapshot per day.
- Minimum 80 s safety buffer between slot start times.
- Checkpoints at baseline, 50, 100 and 150 successful persisted events.

Measured Intelligence Growth



Nine complete measurement sessions, Region 1 (S1 = earliest)

30.6% -> 75.6%

overall closed-loop intelligence score

100%

LLM generation reliability

100%

End-to-end detector capture

71.4%

Severity accuracy

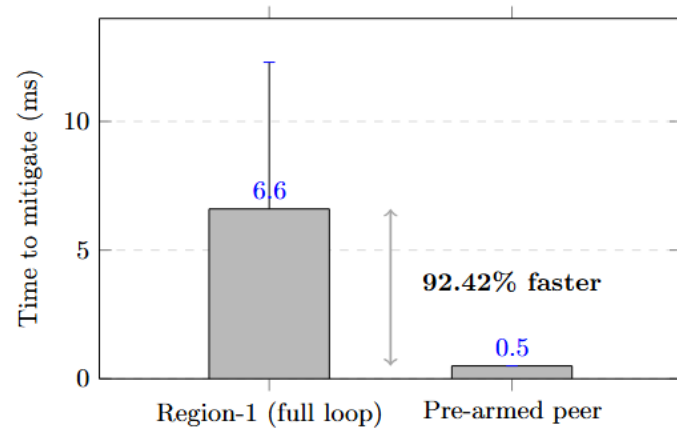
54.2%

Grounded Recall@3

Mean 64.4% · SD 12.6 pp · n = 9

Retrieval grounding and routing remain the main improvement areas.

Automated Mitigation Results



Mean time-to-mitigate over $N = 10$ bounded quarantine runs

A screenshot of the 'Automated Mitigation Lifecycle' operator view. It shows a table with columns: 'Updated', 'Threat', 'Actuator and target', 'Data', and 'State'. The table contains several rows of mitigation events. A specific row is highlighted with a green circle, showing a 'Credential rotation' event for threat 'con-345c74b6ef4c3d6' with state 'executed_unverified'. Below the table, there is a section for 'Credential rotation' with a status bar showing 'Pending', 'Approved', 'Executing', 'Executed', 'Verifying', and 'Verified'. The 'Executing' status is currently active.

Updated	Threat	Actuator and target	Data	State
8/3/2025, 8:12:20 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified
8/3/2025, 8:43:00 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified
8/3/2025, 8:54:20 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified
8/3/2025, 9:36:20 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified
8/3/2025, 9:57:00 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified
8/3/2025, 10:07:00 PM	con-345c74b6ef4c3d6	Credential rotation	Secret: demo-managed-credential-key-token	executed_unverified

Persisted candidate lifecycle in the operator view

92.42%

faster on a pre-armed peer

6.6 ms full regional loop vs 0.5 ms when the peer already holds the intelligence.

5

allow-listed actuators

E2SM-RC throttle, UE access revocation, pod quarantine, credential rotation and A1 policy.

0.85

minimum confidence gate

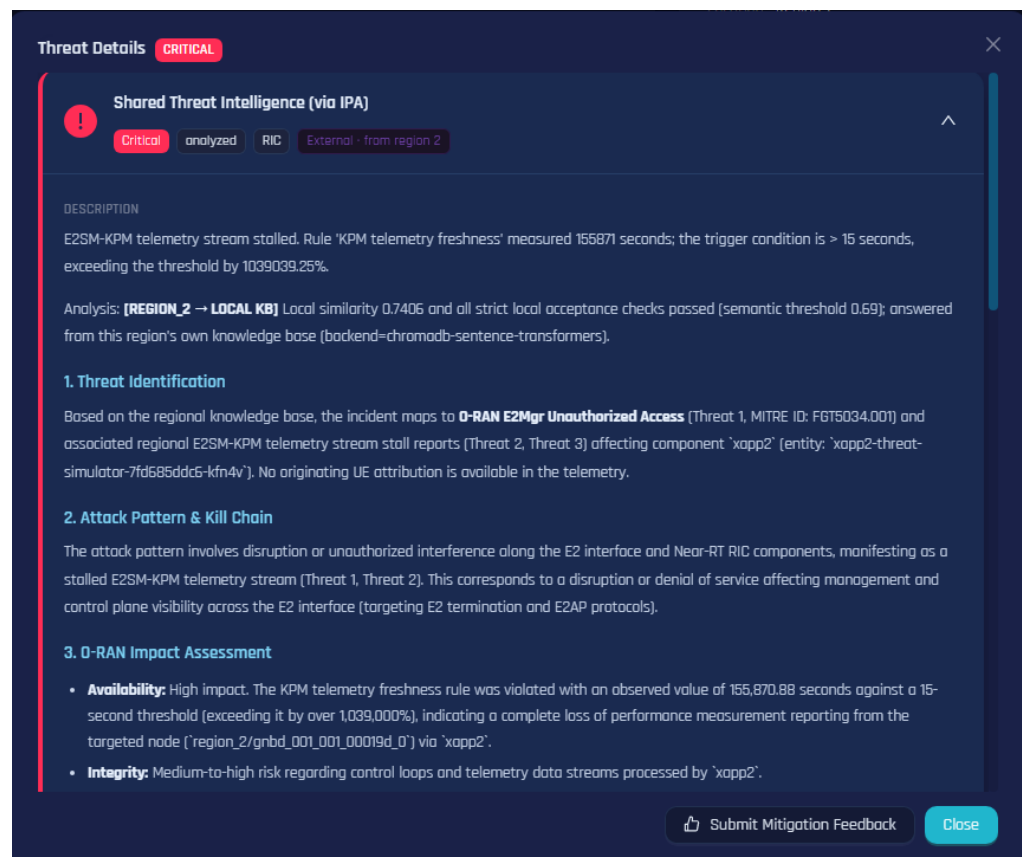
Below it, or on a protected target, the candidate becomes an operator recommendation only.

Rollback

on verification failure

executed_unverified is recorded distinctly and never counts towards mitigation success.

Inter-Platform Sharing Results



The screenshot displays a 'Threat Details' window with a 'CRITICAL' status. The title is 'Shared Threat Intelligence (via IPA)'. Below the title, there are tags: 'Critical', 'analyzed', 'RIC', and 'External - from region 2'. The main content area is divided into sections: 'DESCRIPTION', 'Analysis', '1. Threat Identification', '2. Attack Pattern & Kill Chain', and '3. O-RAN Impact Assessment'. The 'DESCRIPTION' section states: 'E2SM-KPM telemetry stream stalled. Rule 'KPM telemetry freshness' measured 155871 seconds; the trigger condition is > 15 seconds, exceeding the threshold by 1039039.25%.' The 'Analysis' section states: 'Analysis: [REGION_2 -> LOCAL KB] Local similarity 0.7406 and all strict local acceptance checks passed (semantic threshold 0.69); answered from this region's own knowledge base (backend=chromadb-sentence-transformers).' The '1. Threat Identification' section states: 'Based on the regional knowledge base, the incident maps to O-RAN E2Mgr Unauthorized Access (Threat 1, MITRE ID: FGT5034.001) and associated regional E2SM-KPM telemetry stream stall reports (Threat 2, Threat 3) affecting component 'xapp2' (entity: 'xapp2-threat-simulator-7fd585ddc5-kfn4v'). No originating UE attribution is available in the telemetry.' The '2. Attack Pattern & Kill Chain' section states: 'The attack pattern involves disruption or unauthorized interference along the E2 interface and Near-RT RIC components, manifesting as a stalled E2SM-KPM telemetry stream (Threat 1, Threat 2). This corresponds to a disruption or denial of service affecting management and control plane visibility across the E2 interface (targeting E2 termination and E2AP protocols).' The '3. O-RAN Impact Assessment' section contains two bullet points: 'Availability: High impact. The KPM telemetry freshness rule was violated with an observed value of 155,870.88 seconds against a 15-second threshold (exceeding it by over 1,039,000%), indicating a complete loss of performance measurement reporting from the targeted node ('region_2/gnbd_001_0010019d_0') via 'xapp2'.' and 'Integrity: Medium-to-high risk regarding control loops and telemetry data streams processed by 'xapp2'.' At the bottom of the window, there are buttons for 'Submit Mitigation Feedback' and 'Close'.

Threat intelligence received through the Inter-Platform Agent, marked external with its source region



Source exclusion verified

The originating region is absent from every distribution target set, so intelligence is never reflected back to its author.



Peers correctly deduplicated

Peers that already held an equivalent record returned already_present with a similarity score and were skipped rather than re-ingested.



Provenance preserved

Destination records carry external and source-region metadata, so an operator can always distinguish shared intel from a local detection.



Failure isolated, not fatal

An unreachable peer is skipped and logged; the remaining regions still receive the intelligence.



Threshold is a trade-off

Too low suppresses useful contextual variants, too high admits duplicates, so every decision retains its similarity score and reasoning for audit.

Benchmark Comparison

Six ground-truth 5G security scenarios executed concurrently against our Global-RAG platform and an isolated OpenCTI baseline, the leading open-source enterprise CTI platform.

Evaluation metric	Our platform	OpenCTI 6.0+	Observation
STIX precision (Recall@3)	100.0%	0.0%	Complete 5G retrieval superiority
STIX taxonomy alignment	100.0%	33.3%	Native MITRE FiGHT 5G encoding
End-to-end detection	100.0%	100.0%	Equal event capture
Platform stability	100.0%	100.0%	Equal high availability
STIX confidence validation	90.0%	100.0%	Dynamic RAG vs. static schema
Operational CTI latency	3755 ms	37.7 ms	Deep reasoning vs. keyword lookup

Why OpenCTI scores 0% on precision

It runs static keyword lookups over enterprise IT feeds that contain no 5G RAN context to retrieve.

The honest trade-off

We are roughly 100x slower per query. That is the cost of vector embedding, MCP escalation and generative synthesis, and it buys domain precision a keyword index cannot reach.

Objectives Met & Declared Limitations

Objective completion

O1	Disaggregated O-RAN CTI platform implemented across four regions, with O-RAN and cloud-native evidence, dynamic registries and operator visibility.	Implemented
O2	Agentic AI integrated through regional RAG, Global-RAG and MCP for hierarchical contextualisation, controlled learning and safety-gated response.	Implemented
O3	Validated with a frozen benchmark, controlled detector and mitigation experiments, an OpenCTI comparison and regional security-score observation.	Evaluated
E1	Anomaly Detection Gateway delivered: four evidence sources, deterministic scoring, immutable ThreatEvent ledger.	Implemented
E2	Automated mitigation pipeline delivered: five actuators, safety gates, PromQL verification and rollback.	Implemented

Declared limitations

Single-node K3s cluster: the platform is logically, not physically, distributed. No real inter-region latency or partitions.
A1, F1 and selected E2 anomalies are detector-contract simulations, permanently flagged simulation=true.
No multi-vendor or commercial hardware validation; srsRAN and Open5GS only.
Score weights are pre-declared, not empirically calibrated against an external ground truth.
Per-UE containment is realised in the 5G core, because srsRAN answers an E2SM-RC control request with RIC Control Failure.
Cross-region mitigation advantage measured over N = 10 runs, treated as indicative rather than a universal claim.

Future Work

Deploy across physical nodes

Measure real inter-region latency, partitions and independent failure domains.

Use local / paid resilient models

Reduce quota dependence, migrate deprecated SDKs and evaluate provider diversity.

Expand frozen ground truth

Add reviewed O-RAN cases, adversarial paraphrases and external assessors.

Improve retrieval quality

Hybrid retrieval, reranking, metadata filters and validation of learned knowledge.

Strengthen control verification

Broader E2SM-RC profiles, causal effect checks and richer rollback policies.

Longitudinal evaluation

Run multi-week campaigns and test whether local resolution rises without quality drift.

Thank you

Questions and discussion

4

O-RAN regions

750

emulated events

75.6%

closed-loop score

100%

vs 0% STIX precision